

Simone Carletti

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About Me

I am a **Computer Engineering M.Sc.** student specializing in **Robotics and Autonomous Systems**. I'm currently completing my master's thesis at **EPFL**, where I am developing a control system for a **morphing-wing drone** using **Reinforcement Learning** and **$\mathcal{L}_1$ Adaptive Control**. I have hands-on experience across the robotics stack, from **perception** and **planning** to **control systems** and **embedded hardware**, with proficiency in **C++**, **Python**, **ROS2**, and **TypeScript**. I am experienced in transitioning algorithms from Python/Simulink simulations to high-performance embedded implementations (C/Rust). Beyond my research at **EPFL** and **DTU**, my **professional experience** as a **lead developer** has taught me to write clean, maintainable, and production-ready code.

Education

EPFL - LIS, Master's Thesis - Title: L1-Adaptive augmentation for robust morphing-wing drone flight control - Skills: Control Systems, RL (PPO), C++, Python, Simulink, ROS, Git, Linux, Unity	<i>Sept 2025 – Present</i>
DTU, M.Sc. Autonomous Systems Erasmus+ exchange - Converted GPA: 30/30	<i>Sept 2024 – Jul 2025</i>
Politecnico di Torino, M.Sc. Computer Engineering - Graduated with 110/110 and honors (GPA: 29.48/30) - Curriculum in Automation and Intelligent Cyber-Physical Systems	<i>Sept 2023 – Present</i>
Università degli Studi di Firenze, B.Sc. Computer Science and Engineering - Graduated with 110/110 and honors - Thesis: Sensor network for monitoring the greenhouses of the Botanical Garden of Florence	<i>Sept 2020 – Sept 2023</i>

Experience

Robotics Engineer, DIANA student team Collaborated on a Mars rover student project, implementing perception and EKF state estimation in C++ while contributing to software development, testing, and system integration	<i>Sept 2023 – Jul 2024 Turin, Italy</i>
Full-Stack Developer, Genomedics S.r.l - Lead developer of three key projects that handle the release, deployment, and licence management for all the company's products. Developer for the company's main project (KPlatform) - Skills: Angular (Typescript, SCSS, HTML), Vue.js, PHP, Laravel, SQL, Python, Git, Linux	<i>Sept 2020 – Present Florence, Italy</i>

Selected Projects

Autonomous Navigation & Task Execution - DTU RoboCup 🏆 Developed autonomous navigation, control, and task execution algorithms with continuous integration and validation tests on the real platform	<i>Feb 2025 – May 2025</i>
Digital Control - Autonomous Robotics (GitHub 📄 - git.new/dtudc) Designed digital controllers and implemented them on embedded hardware using C++	<i>Oct 2024 – Dec 2024</i>
3D Object Tracking using Stereo Camera (GitHub 📄 - git.new/dtup) Developed a 3D perception pipeline integrating Faster R-CNN for detection, SIFT-based stereo imaging, and EKF for robust object tracking	<i>Nov 2024 – Dec 2024</i>
Comparison and implementation (from scratch) of YOLOv1 and U-Net Implemented landmark computer vision models from scratch using PyTorch (GitHub 📄)	<i>Oct 2024 – Dec 2024</i>
Multiplayer videogame on embedded hardware (GitHub 📄 - git.new/quor) Developed bare-metal C firmware on ARM Cortex-M3 with a custom CAN protocol	<i>Jan 2024 – Jun 2024</i>
Localization with EKF in ROS2 Implemented a full EKF localization pipeline from scratch in ROS2 and Python, fusing data from LiDAR, IMU, and wheel odometry	<i>Nov 2023 – Dec 2023</i>

Skills & Activities

- Languages:** Fluent in English, Native Italian speaker, Basic Spanish skills
- Programming:** C++, C, Python, TypeScript, JavaScript, Rust, Java, Matlab, PHP, SQL, HTML, CSS/SCSS, Assembly, LaTeX
- Tools:** PyTorch, Linux, Git, Google Colab, OpenCV, Onshape, CMake, Docker, Node-RED, Multisim
- Other:** Raspberry Pi, Arduino, ESP8266, NodeMCU, Soldering, Nvidia Jetson/Orin, Guitar, Photography, Networking, Zigbee
- Volunteering:** ESN Copenhagen and DTU (Organized events, supported international students and promoted cultural exchange)